

REFERENCE DESIGN

January 10, 2025

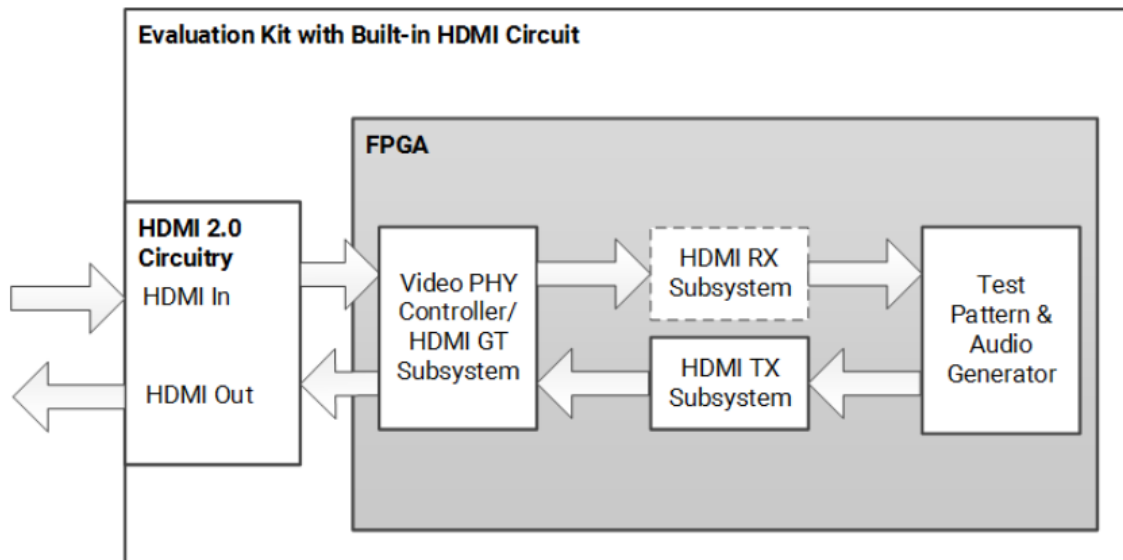
Reference Design:

AMD HDMI Pass Through Reference Design targeting TRIA AUBoard-15P Development Kit using 2023.1 tool flow.

Introduction:

This reference design demonstrates passing video through HDMI connections on the AUBoard-15P Development Kit.

- Supports HDMI Transmit / Receive resolutions up to 4K60



Webpage:

<http://avnet.me/auboard-15p-dk-hdmi>

Reference Documents:

[HDMI Intellectual Property IP Landing Page \(EF-DI-HDMI-SITE\)](#)

[HDMI 1.4/2.0 Receiver Subsystem Product Guide \(PG236\)](#)

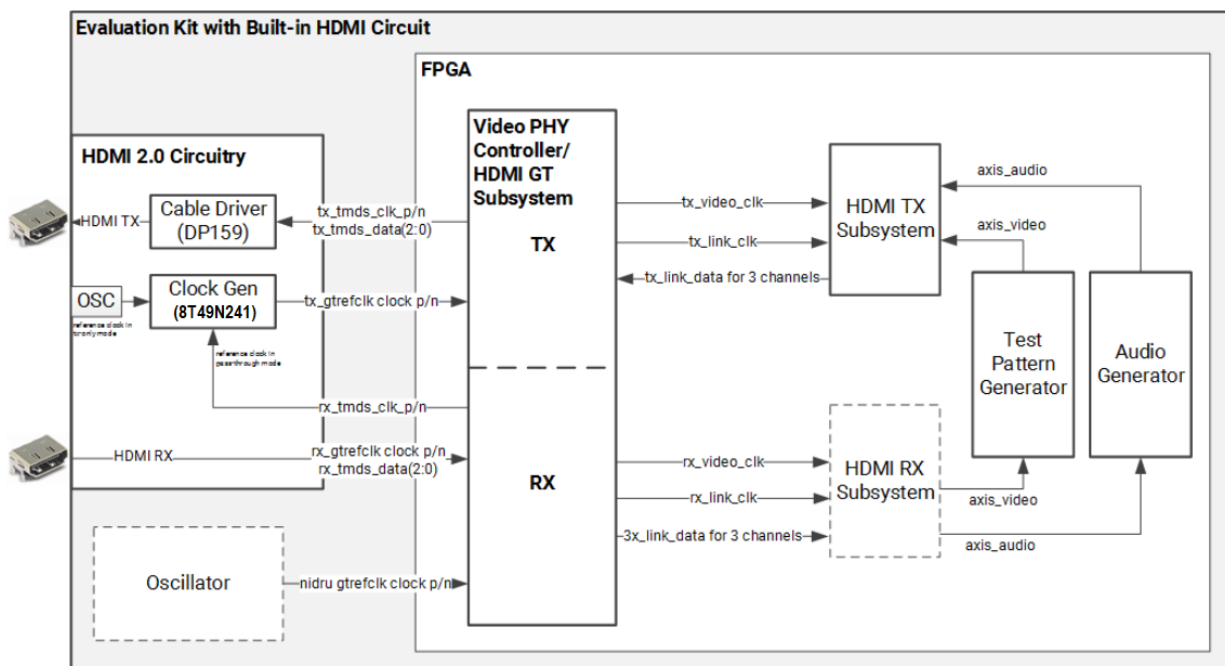
[HDMI 1.4/2.0 Transmitter Subsystem Product Guide \(PG235\)](#)

Pertinent Information:

This reference design is generated from the step-by-step instructions listed in the HDMI Subsystem Product Guide documents referenced above. While those instructions target Development Boards from AMD, they have been modified to be able to target the AUBoard-15P.

The AUBoard-15P is an FPGA board with on-board HDMI TX/RX. It is most like the KC705/KCU104/VCU118 reference designs with the exception that the clock generator chip used in the HDMI circuit targets an 8T49N241 device. The provided source files contain the modifications for the retarget to the AUBoard-15P.

This bare metal reference design can be tested by running the design through the Vitis 2023.1 for either the pass-through or transmit reference designs. The design is setup / monitored through a terminal as listed in the provided documents.

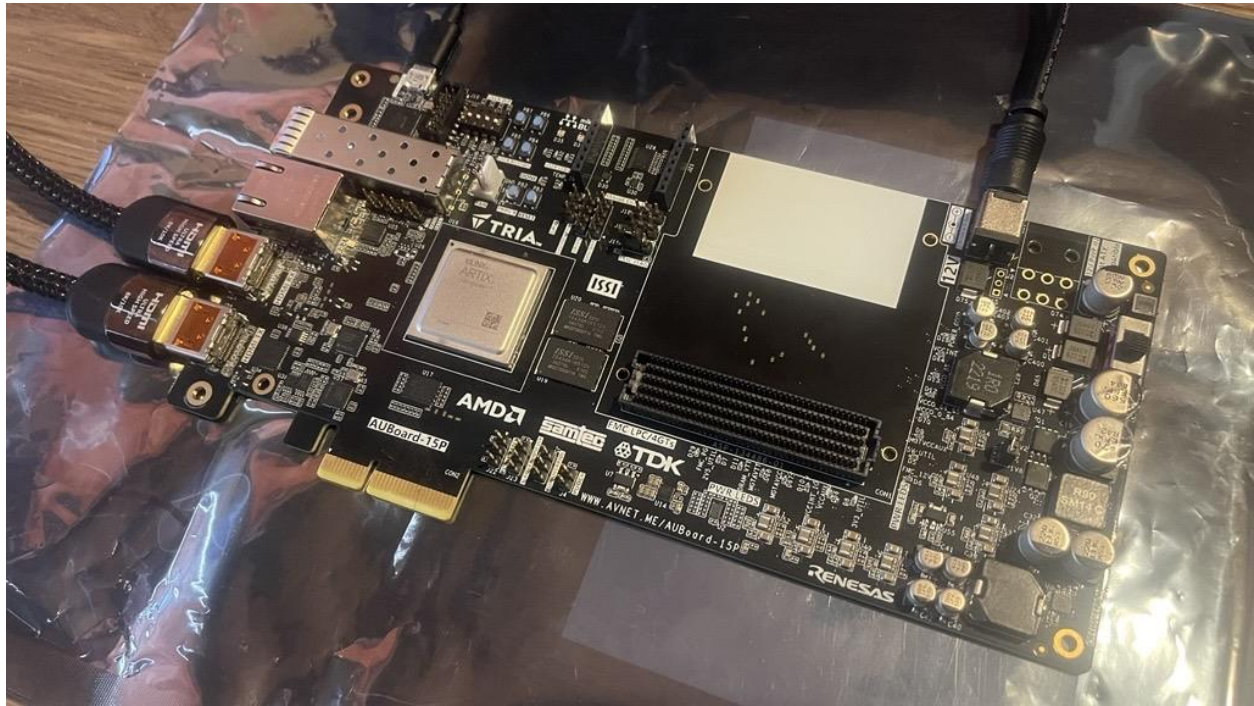


The HDMI Pass Through Reference Design is tested with the following sink devices:

- Samsung UHDTV/HDTV

Equipment Required:

- HDMI Cables to support 4K60 resolution – recommend thin connector jackets.
- MicroUSB cable for JTAG/Terminal Debug support
- 12V Power Supply
- AUBoard-15P
- Suitable HDMI Transmit Source – Example: HDMI laptop output.
- Suitable HDMI Receive Source – Understand monitor HDMI specific ports resolution/frame rate support.



Additional Support:

For additional support, please review the discussions and post your questions in the AUBOARD-15P Development Kit Forum located here: <http://avnet.me/AUBOARD-15P-forum>

Alternatively, you can also reach out to your local Avnet Field Application Engineer (FAE) for support.

Revision History:

Date	Version	Revision
10-Jan-25	1.0	Initial Release